

Real-Time Flood Risk Prediction

Sunamganj Haor — RF + XGBoost | 11 Active Features | LOOCV 89.6% (real-SAR, deconfounded)

Refresh Live Data

LSTM skipped (feature mismatch — retrain LSTM to fix): X has 10 features, but MinMaxScaler is expecting 11 features as input.

⚠️ UPSTREAM FLOOD SIGNAL — Barak River (Assam/India)

বাংলা: উজানের সংকেত — বরাক নদীতে পানি বেড়েছে (~36 ঘণ্টার মধ্যে আওরে পৌঁছাতে পারে)

Upstream VV: -8.2 dB | Estimated arrival: ~36 hours

This may indicate barrage/gate release. Prepare NOW — এখনই প্রস্তুত হন।

Risk Level

EXTREME

↑ অত্যন্ত বিপদজনক

Flood Probability

87.4%

↑ 95% CI: 74–100%

Inundated Area

122.4 km²

Forecast 12h

68.5 mm

Forecast 72h

177.2 mm

Est. Flood Duration

49 days

Model Technical Details — RF + XGBoost + LSTM

Model Breakdown

Random Forest

94.4%

500 trees · 10 features · LOOCV 89.6% (real-SAR) · 87.8% (extended, deconfounded)

XGBoost

99.7%

500 estimators · 10 features · LOOCV 89.6% (real-SAR) · 87.8% (extended, deconfounded)

LSTM (5-day sequence)

50.0%

⚠️ Synthetic training · walk-forward 100% (overfit, n=101) · excluded from primary metric

Ensemble: RF×0.45 + XGB×0.35 + LSTM×0.2 = 87.4% (95% CI: 74%–100%)

⚠️ High model uncertainty — 95% CI spans 26 percentage points (74%–100%). Verify with FFWC real-time gauge data before issuing community alerts. মডেলের অনিশ্চয়তা বেশি — FFWC গেজ তথ্য দিয়ে যাচাই করুন।

RF Uncertainty Detail — Tree Probability Distribution

RF Forest: Distribution of 500 tree predictions

Each bar = number of trees predicting that probability. Narrow distribution = high model confidence. RF std = 15.5pp | 95% CI = ±13.7pp

5-Day Rainfall Forecast (Sunamganj Haor)

5-Day Flood Risk Forecast — Sunamganj Haor

07-10

EXTREME

27.9 mm

07-11

EXTREME

89.0 mm

07-12

EXTREME

35.7 mm

07-13

EXTREME

49.4 mm

07-14

EXTREME

29.5 mm

72-Hour Flood Risk Forecast

Last updated: 2026-07-10 22:29 BST (cached 1 h) · Open-Meteo hourly precipitation → RF+XGB ensemble

PEAK 72-HOUR FLOOD PROBABILITY

95.0% EXTREME RISK

Peak window: 0–24h

Total 72h Forecast Rain

177.2 mm

↑ open-meteo

Data source

Open-Meteo

↑ Hourly · Dhaka TZ

আগামী ৭২ ঘণ্টায় অত্যন্ত বিপদজনক বন্যার ঝুঁকি রয়েছে। এখনই নিরাপদ স্থানে যাওয়ার প্রস্তুতি নিন। ফসল ও গবাদিপশু সরিয়ে ফেলুন।

72-Hour Flood Risk by Window — Sunamganj Haor

0-24h

95.0%

↑ 83.7 mm forecast

Risk: EXTREME

24-48h

95.0%

↑ 30.8 mm forecast

Risk: EXTREME

48-72h

95.0%

↑ 62.7 mm forecast

Risk: EXTREME

Hourly Rainfall Breakdown (72h)

Hourly Forecast Precipitation — Sunamganj Haor (next 72h)

Source: open-meteo · Fetched: 2026-07-10 15:48 UTC · 72h total: 177.2 mm

Prediction Component Breakdown

Shows how each layer contributes to the peak 72-hour flood probability. Layer 1 = ML satellite/rainfall base · Layer 2 = current discharge level · Layer 3 = discharge rising trend (reliability-gated).

Layer 3 (discharge trend) suppressed — Trend unreliable: [slope] = 14 < 100 m³/s/day (within noise level). Layer 3 adjustment suppressed. Only Layers 1 (ML) and 2 (discharge level) contribute to this forecast. Raw slope: +14 m³/s/day · R² = 0.94 · Days: 15

FINAL FLOOD PROBABILITY — PEAK WINDOW (0–24H)

95.0%

Component Breakdown:

ML base (satellite + rainfall) : 96.2%

Current discharge (Layer 2) : +0.0 pp

Discharge trend (Layer 3) : +0.0 pp

TOTAL = 96.2 + 0.0 + 0.0 = 95.0%

Safety cap applied: L2+L3 combined adjustment limited to 30 pp and/or final probability capped at 95% to prevent discharge from overpowering the ML signal.

Explanation: Satellite and rainfall conditions indicate 96% probability. Upstream discharge is normal — ML estimate is the primary signal (Layer 3 suppressed — trend unreliable).

Per-Window 3-Layer Contribution Chart

3-Layer Flood Probability — Component Contribution per Window

Window	Base (L1) %	+Discharge (L2)	+Trend (L3)	Final %	L2 Reason	L3 Reason
0-24h	96.2	+0.0 pp	+0.0 pp	95.0	Normal/rising discharge (795 m ³ /s) → no L2 adjustment	Layer 3 suppressed — trend unreliable (R ² < 0.60 or [slope] < 100 m ³ /s/day)
24-48h	96.7	+0.0 pp	+0.0 pp	95.0	Normal/rising discharge (795 m ³ /s) → no L2 adjustment	Layer 3 suppressed — trend unreliable (R ² < 0.60 or [slope] < 100 m ³ /s/day)
48-72h	96.7	+0.0 pp	+0.0 pp	95.0	Normal/rising discharge (795 m ³ /s) → no L2 adjustment	Layer 3 suppressed — trend unreliable (R ² < 0.60 or [slope] < 100 m ³ /s/day)

14-Day Discharge History + Trend Line

Barak Discharge — 14-Day History + 3-Day Smoothing + 72h Projection

Current Discharge

795 m³/s

↑ GloFAS reanalysis (Open-Meteo Flood API)

Raw Trend (OLS)

+14 m³/s/day

⚠️ Trend suppressed — Trend unreliable: [slope] = 14 < 100 m³/s/day (within noise level). Layer 3 adjustment suppressed. 72h projection shows flat line (795 m³/s) — no trend extrapolation applied.

Trend R²

0.94

⚠️ Suppressed (<0.60)

72h Projected

795 m³/s

↑ +0 vs now

Upstream Barak River Discharge Monitoring

Barak at Silchar, Assam (24.82°N 92.79°E) · GloFAS reanalysis via Open-Meteo · 72h projection from 14-day trend · Last updated: 2026-07-10 22:29 BST (cached 1h)

CURRENT BARAK DISCHARGE (SILCHAR)

704 m³/s Normal

Normal discharge (704 m³/s) — no upstream flood signal.

Trend

Stable

↑ +95 m³/s/day

ML Base Prob

87.4%

↑ RF+XGB ensemble

Discharge-Adjusted Prob

87.4%

↑ No adjustment

Barak River Discharge — Historical (14d) + 72h Trend Projection

0-24h

799 m³/s

↑ Normal

24-48h

893 m³/s

↑ Normal

48-72h

988 m³/s

↑ Normal

Data: trend_projection · 72-hour discharge projected from 10-day GloFAS trend. Not a direct GloFAS ensemble forecast — treat as indicative only. · Fetched: 2026-07-10 15:48 UTC

Surma River Gauge — Sunamganj (Direct Haor Hydraulic Driver)

GloFAS reanalysis at Sunamganj (24.87°N, 91.40°E) · Open-Meteo Flood API · High Barak discharge (upstream) typically causes high Surma discharge 6–12h later. Both signals together give the most complete haor flood picture. Source: GloFAS reanalysis via Open-Meteo Flood API (Sunamganj 24.87°N, 91.40°E). Excluded from ML model due to multicollinearity — not a model weakness, but a deliberate feature engineering decision to prevent overfitting.

CURRENT SURMA DISCHARGE (SUNAMGANJ)

75.3 m³/s Normal

Normal discharge (75 m³/s) — no local flood signal.

% of Danger Level

9%

↑ Danger = 800 m³/s

ML Model Role

Dashboar...

↑ r=0.79 multicollinear

Approx. Cusecs

2 cusec

↑ FFWC reporting unit

Surma Discharge Thresholds (BWDB Sunamganj): Normal < 150 · Rising 150–400 · High 400–800 · Danger ≥ 800 m³/s

Why Surma matters: Surma discharge directly drives haor water level at Sunamganj. High Barak discharge (upstream) typically causes high Surma discharge 6–12h later. Both signals together give the most complete haor flood picture. Source: GloFAS reanalysis via Open-Meteo Flood API (Sunamganj 24.87°N, 91.40°E). Excluded from ML model due to multicollinearity — not a model weakness, but a deliberate feature engineering decision to prevent overfitting.

All 15 Live Features — Raw Sensor Values

All 15 Live Features (13 ML inputs + Surma + Barak discharge)

SAR + Terrain

Meteorological

New Features

Forecast

Sentinel-1 VV

-20.31 dB

Slope (SRTM)

1.91°

Sentinel-1 VH

-27.74 dB

Data window

2026-07-01 → 2026-07-08

VV/VH Ratio

0.7320

Data: Sentinel-1 GRD · Sentinel-2 · CHIRPS · ERAS-Land · HydroSHEDS · Open-Meteo · Barak upstream proxy | 13 features fetched | 10 used by RF/XGB | Cached 1h